BIRATIONALITY OF IRREDUCIBLE SCHEMES IS EQUIVALENT TO HAVING ISOMORPHIC NONEMPTY OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. The relative (over a base scheme) part of the statement is dispatched by the author with 'In the S-rational maps case, the exact same argument works.' – a self-described repetition of the argument just completed, accepted with the wording quoted under the Lead's 12:4x rule, not treated as an omitted statement item.

Allowed standard prerequisites (editorial).

- A section of a morphism whose composite is the identity is an immersion. Schemes, Lemma \ref{schemes-lemma-section-immersion}, cited at proof lines 12919-12921. Shows the morphism between the chosen opens is an immersion.
- Immersions are monomorphisms. Schemes, Lemma \ref{schemes-lemma-immersions-monomorphisms}, cited at proof lines 12925-12927. Turns a monomorphism which is an immersion into an open immersion.
- Dominant rational maps and their composition (the definition of birationality). Definition \ref{definition-birational}, the definition this lemma works with. Supplies the inverse rational maps the proof starts from.

Source reference. Original lemma 0BAA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14203–14203. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01KT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2062–2062. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01L7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2076–2076. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let X and Y be irreducible schemes.*

(1) *The schemes X and Y are birational if and only if they have isomorphic nonempty opens.*
(2) *Assume X and Y are schemes over a base scheme S. Then X and Y are S-birational if and only if there are nonempty opens $U \subset X$ and $V \subset Y$ which are S-isomorphic.*

Proof. Assume X and Y are birational. Let $f : U \to Y$ and $g : V \to X$ define inverse dominant rational maps from X to Y and from Y to X. We may assume V affine. We may replace U by an affine open of $f^{-1}(V)$. As $g \circ f$ is the identity as a dominant rational map, we see that the composition $U \to V \to X$ is the identity on a dense open of U. Thus after replacing U by a smaller affine open we may assume that $U \to V \to X$ is the inclusion of U into X. It follows that $U \to V$ is an immersion (apply Schemes, Lemma 01KT to $U \to g^{-1}(U) \to U$). However, switching the roles of U and V and redoing the argument above, we see that there exists a nonempty affine open $V' \subset V$ such that the inclusion factors as $V' \to U \to V$. Then $V' \to U$ is necessarily an open immersion. Namely, $V' \to f^{-1}(V') \to V'$ are monomorphisms (Schemes, Lemma 01L7) composing to the identity, hence isomorphisms. Thus V' is isomorphic to an open of both X and Y. In the S-rational maps case, the exact same argument works. □